UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

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SECJ1023 PROGRAMMING TECHNIQUE II

SECTION 02

PROJECT 1: PROJECT PROPOSAL

SELF-CHECKOUT SYSTEM

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GROUP 4

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Introduction

A self-checkout system is a technologically advanced solution that provides benefits not only to customers but also to management. For example, it allows customers to scan, bag, and pay for their products without the assistance of a cashier. Because of this, customers no longer need to wait in line to pay for their products. This is especially useful for those who only want to buy a few things and don't want to wait in line behind others that purchase a large number of items. The majority of payment methods, especially credit cards, can be completed in a matter of seconds using this technology. On the other hand, through the self-checkout system, management can reduce part of the store's cost of hiring cashiers, save the store a sum of money to improve the shopping environment for customers in the store, and attract and retain more customers.

Synopsis and General Idea

The purpose of this proposal is to suggest the development of an OOP-based Self-Checkout System, that implements OOP's principles such as encapsulation, inheritance, polymorphism, and abstraction. This system, which will be designed in C++ programming environment, will provide multiple functionalities to the users, not only to customers but also to the management. The system will include management and customer classes to help organize the checkout process in the store. It allows customers to know the details of their shopping by printing the receipt for them, and also assists management to better manage the financial status of the store and gain a better understanding of the store's profitability by providing them with a summary of the store's financial health.

In the proposed system, customers can complete the checkout process without the assistance of a cashier, and after checkout they can clearly see what they bought, how much tax was charged, whether there are discounts and current membership points on the receipt. For management, self-checkout systems are even more helpful. Management can easily conduct transaction data analysis. Transaction data includes information about the items purchased, the time and date of the transaction, the payment method used and any discounts applied. From this, managers can use transaction data to identify popular products, purchasing trends, and peak shopping periods. For example, checking the current sales figures allows management to decide whether to increase the purchase volume of the product, and also allows management to find an appropriate time to carry out promotion to achieve the best publicity and sales results. Additionally, transaction data can be used to improve inventory management processes. Managers can detect slow-moving items, better estimate demand and modify inventory levels by observing individual product sales in real time. This prevents overstocking or stockouts, thereby increasing inventory turns and reducing costs.

From the perspective of the customer, the checkout process begins with the entry of the member ID (if applicable). Once the ID has been located, the screen will display the customer's name and member point and the process will proceed to enter the code of the items. The total price, discount, tax, and member points will all be automatically calculated by the system. After that, the user has the option to review the products that has purchased again and enter “-1” to proceed with the payment. The machine will then provide the customer with a receipt once the payment has been completed.

From the viewpoint of managers, the system makes it easier for them to analyze data. For a summary of each day's sales, the total number of products sold, the total number of transactions, the average value per transaction, and the most popular item sold that day, key in "1". By entering "2", he can view member details, including name, member ID, and respective member points. Finally, entering "3" will terminate the program.

Expected Output

Customer View with Example Input Shown in Bold

```
**************************************************
            The Perfect Grocery Shop
**************************************************
Press any key to start... [Enter]
Member ID (Enter -1 if not a member): 4678 [Enter]

Name:     Adam bin Adam
Points:   285

Enter the item code: 2358 [Enter]

Code  Product Name                 Unit Price(RM)
2358  MASSIMO WHEAT BREAD          4.90
Subtotal                           4.90

Enter -1 to proceed to payment
Enter the item code: 1384 [Enter]

Code  Product Name                 Unit Price(RM)
2358  MASSIMO WHEAT BREAD          4.90
1384  BUNTONG GINGER               7.90
Subtotal                      12.80

Enter -1 to proceed to payment
Enter the item code: 9980 [Enter]

Item not found.

Enter -1 to proceed to payment
Enter the item code: 1248 [Enter]

Code  Product Name                 Unit Price(RM)
2358  MASSIMO WHEAT BREAD          4.90
1384  BUNTONG GINGER               7.90
1248  STABILO PENCIL 12PSC         16.99
Subtotal                           29.79

Enter -1 to proceed to payment
Enter the item code: -1 [Enter]
```

```
**************************************************
          The Perfect Grocery Shop
         Jalan Computing, Taman UTM,
          Skudai, 12234 Johor Bahru
               No.Tel: 051233523
       Email: support@perfectgrocery.com
**************************************************
Code  Product Name                  Unit Price(RM)
2358  MASSIMO WHEAT BREAD           4.90
1384  BUNTONG GINGER                7.90
1248  STABILO PENCIL 12PSC          16.99

Item Count                          3
Subtotal                            29.79
Discount                            0.00
SST(10%)                            2.98
Rounding                            0.01
Member points                       314

**************************************************
            Thank you and come again
**************************************************
```

Manager View with Example Input Shown in Bold

```
**************************************************
            The Perfect Grocery Shop
**************************************************
```

1. Display Current Day Sales
2. Display Member Information
3. Quit Program

Choice: **1[Enter]**

```
**************************************************
            The Perfect Grocery Shop
**************************************************
Total Sales: RM 135615.35
Total Items Sold: 13589 items
Total Transactions: 1083
Average Amount per Transaction: RM 125.22
Most popular item of the day: 2358 MASSIMO WHEAT BREAD
```

```
**************************************************
            The Perfect Grocery Shop
**************************************************
   1. Display Current Day Sales
   2. Display Member Information
   3. Quit Program
Choice: 2[Enter]

**************************************************
            The Perfect Grocery Shop
**************************************************
Name                     ID             Points
Ali bin Ali              4678           314
Abu bin Abu              4679           129
Lim Zhi Ping             4680           129
Prashanth A/L Pravinn    4681           195
Siti binti Yunus         4682           481
Jane Wong Mei Kei        4683           357

**************************************************
            The Perfect Grocery Shop
**************************************************
   1. Display Current Day Sales
   2. Display Member Information
   3. Quit Program
Choice: 3[Enter]
```

Storyboard


SELF CHECKOUT
WELCOME!
Touch Screen To START
PUT YOUR ITEMS HERE
LONG QUEUE!
3
COUNTER
SELF CHECKOUT
Welcome
Welcome!
Touch Screen to Start
Please scan member card.
Perfect Grocery Store
No, I don't have a member card.
Member card detected.
Member ID : 5019
Point : 310
Continue
Please scan your item
Code Product Price
Total RM
0.00
PAY NOW
BEEP.
Scan your next item or select Pay Now!
Code Product Price
2358 Potatoes 1.99
1364 Onion 3.00
5718 Milk 8.50
6032 Pumpkin 3.00
Total RM
16.49
PAY NOW
PAY NOW
SELECT PAYMENT
E-WALLET
CARD
PLEASE INSERT CARD
PRINTS RECEIPT
PERFECT

Conclusion

In conclusion, the proposed Object-Oriented Programming (OOP)-based Self-Checkout System promises to revolutionize the checkout process by integrating seamlessly designed features. By adhering to OOP principles, our system ensures smooth operation and enhanced efficiency. Through intuitive interfaces and automated functionalities, such as item scanning and financial summary checking, our system aims to optimize cashier productivity and elevate customer satisfaction.